

1. Consider the following type of CFGs where production rules are of the form: $A \rightarrow aB$ or $A \rightarrow \epsilon$ for A, B variables and $a \in \Sigma$ is a terminal. Show that the language generated by this type of grammar is exactly the regular set. (12 marks)
2. Let $b(n) \in \{0, 1\}^*$ denote the binary representation of $n \geq 1$ with leading zeroes omitted. Show that the set $\{b(n)Sb(n+1) \mid n \geq 1\}$ is not a CFL. (15 marks)
3. Show that the set $\{0, 1\}^* \setminus \{xx \mid x \in \{0, 1\}^*\}$ is a context-free language. (15 marks)
4. Show that the set $\{x\#y \mid x \text{ is a substring of } y, \text{ and } y \in \{0, 1\}^*\}$ is not a CFL. (12 marks)
5. Show that if A and B are recursively enumerable, then their concatenation AB is also recursively enumerable. (12 marks)
6. Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA and let $h \in Q$ be a state called its "home". A *synchronizing sequence* for M and h is a string $s \in \Sigma^*$ where $\hat{\delta}(q, s) = h$ for every $q \in Q$. Say that M is *synchronizable* if it has a synchronizing sequence for some state h . Prove that, if M is a k -state synchronizable DFA, then it has a synchronizing sequence of length at most 2^k . (12 marks)
7. Let $B = \{\langle M \rangle \mid M \text{ is a Turing machine that takes more than 500 steps on some input}\}$. Show that the set B is decidable. (11 marks)
8. Consider the CFG $G = (\{S, A, B, C\}, \{a, b\}, P, S)$ where P is given as follows
- $$S \rightarrow AB \mid BC, A \rightarrow BA \mid a, B \rightarrow CC \mid b, C \rightarrow AB \mid a$$
- Prove or disprove that the string $baaba$ is in $L(G)$. (11 marks)